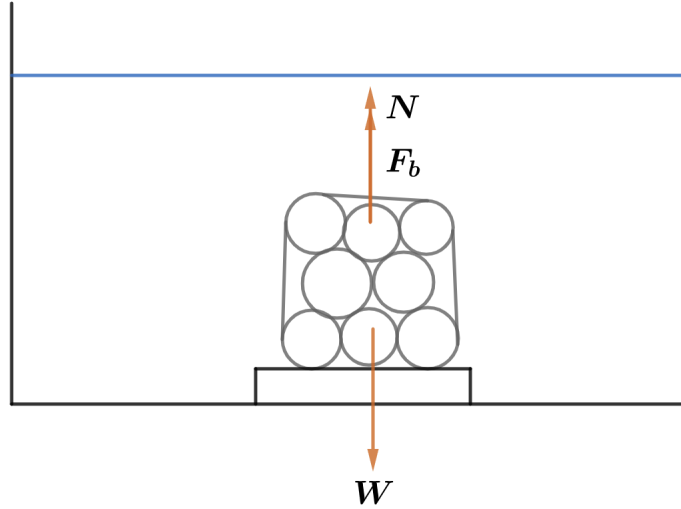


2023 F=ma Exam: Problem 18

Kevin S. Huang



Balancing forces, we have

$$N = W - F_b$$

where N is the normal force (measured by the scale), W is weight of everything inside the bag and F_b is the buoyant force on the bag. We have

$$N_{\text{Alice}} = W_{\text{rocks}} - F_{b,\text{rocks}}$$

$$N_{\text{Bob}} = W_{\text{rocks}} + W_{\text{water}} - F_{b,\text{rocks}} - F_{b,\text{water}} = N_{\text{Alice}}$$

using the fact that $W_{\text{water}} = F_{b,\text{water}}$ by Archimedes' principle.

$$N_{\text{Carol}} = W_{\text{rocks}} + W_{\text{air}} - F_{b,\text{rocks}} - F_{b,\text{air}} < N_{\text{Alice}}$$

since $W_{\text{air}} < F_{b,\text{air}}$ as the density of air is less than the density of water. Thus,

$$W_C < W_A = W_B$$

so the answer is E.